Inverse Cosine Similarity: Algorithm B

Construction, Theory, and Applications

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1 Introduction

The original Algorithm A [1] solves the inverse cosine similarity problem in $\mathcal{O}(n)$ time per query by performing a Gram–Schmidt orthogonalisation at each invocation. In many practical settings — such as real-time embedding perturbation in recommendation engines [4], high-frequency portfolio stress-testing [6], or adversarial example generation in neural networks [3] — the anchor vector $\mathbf{u}$ is *fixed* while the target similarity s varies across thousands or millions of queries. In this regime the per-query Gram–Schmidt step is wasteful.

This paper presents **Algorithm B**, which trades storage for speed by precomputing and caching a full orthonormal basis for $\mathbb{R}^n$ during a one-time $\mathcal{O}(n^2)$ setup phase. Subsequent queries for *any* target $s \in [-1, 1]$ are resolved by a simple two-term linear combination requiring only $\mathcal{O}(n)$ *work* on cached data — with no floating-point divisions, norm computations, or random draws at query time. When the same basis column is reused across queries, the effective amortised cost per query is $\mathcal{O}(1)$ scalar operations after an $\mathcal{O}(n)$ final assembly.

The key contributions are:

1. A precomputed $n \times n$ orthonormal matrix Q whose first column is $\hat{\mathbf{u}}$ (Section 3.1).
2. An $\mathcal{O}(1)$-amortised query procedure that selects a basis column and forms the output by a two-scalar blend (Section 3.2).
3. A batch extension that generates K distinct outputs for K target similarities in $\mathcal{O}(Kn)$ time after the setup phase (Section 3.3).
4. Formal proofs of correctness, termination, and complexity (Section 4).
5. An explicit space–time trade-off characterisation (Section 4.5).

The paper ends with "The End"

2 Background and Notation

2.1 Cosine Similarity

Definition 2.1 (Cosine Similarity)**.** *For* $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$*,*

$$\mathrm{cs}(\mathbf{u}, \mathbf{v}) \;=\; \frac{\mathbf{u}^\top \mathbf{v}}{\|\mathbf{u}\| \, \|\mathbf{v}\|} \;\in\; [-1, 1]. \tag{1}$$

2.2 Orthonormal Basis and QR Decomposition

Definition 2.2 (Orthonormal Basis)**.** *A set* $\{q_1, \ldots, q_n\} \subset \mathbb{R}^n$ *is an* orthonormal basis *(ONB) if* $q_i^\top q_j = \delta_{ij}$ *for all* i, j*, where* δ_{ij} *is the Kronecker delta. Equivalently, the matrix* $Q = [q_1 \mid \cdots \mid q_n] \in \mathbb{R}^{n \times n}$ *satisfies* $Q^\top Q = QQ^\top = I_n$*.*

Definition 2.3 (Reduced QR Decomposition)**.** *For* $\mathbf{u} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$*, the* reduced QR decomposition *of* $\mathbf{u}$ *is the factorisation* $\mathbf{u} = QR$ *where* $Q \in \mathbb{R}^{n \times n}$ *is orthogonal and* $R \in \mathbb{R}^{n \times n}$ *is upper triangular. For our purposes we only need the first column:* $q_1 = \hat{\mathbf{u}} = \mathbf{u}/\|\mathbf{u}\|$*.*

2.3 Householder Reflections

The most numerically stable method for constructing a full ONB with a prescribed first column is via *Householder reflections* [2]. Given $\hat{\mathbf{u}} \in \mathbb{R}^n$, define

$$\mathbf{a} = e_1 \in \mathbb{R}^n, \qquad \mathbf{x} = \hat{\mathbf{u}} - e_1, \qquad H = I - \frac{2\mathbf{x}\mathbf{x}^\top}{\|\mathbf{x}\|^2}, \tag{2}$$

so that $He_1 = \hat{\mathbf{u}}$. The columns of $H^\top = H$ (since H is symmetric and orthogonal) then form an ONB with $q_1 = \hat{\mathbf{u}}$. The full matrix Q of Algorithm B is constructed as $Q = H$ (or a product of Householder reflectors when $\hat{\mathbf{u}} = e_1$ exactly).

3 The Inverse Cosine Similarity Algorithm B: Precomputed Basis Method

Algorithm B has two distinct phases: a **setup phase** executed once per anchor vector, and a **query phase** executed once per target similarity.

3.1 Setup Phase: Precomputing the ONB

Algorithm 1 Algorithm B — Setup Phase

Require: $\mathbf{u} \in \mathbb{R}^n$, $\mathbf{u} \neq \mathbf{0}$, $n \geq 2$
Ensure: Orthonormal matrix $Q \in \mathbb{R}^{n \times n}$ with $q_1 = \hat{\mathbf{u}}$

```
û ← u/‖u‖                                   ▷ O(n)
if ‖û − e_1‖ < ε then
    Q ← [e_2 | e_1 | e_3 | ··· | e_n]       ▷ Swap to avoid degenerate Householder
else
    x ← û − e_1                             ▷ Householder vector, O(n)
    τ ← 2/(x^⊤ x)                           ▷ Scalar, O(n)
    Store (x, τ) implicitly or expand:
    Q ← I_n − τ x x^⊤                       ▷ O(n^2) time, O(n^2) space
end if
return Q
```

Remark 3.1. *The matrix* Q *may be stored* implicitly *as the pair* $(\mathbf{x}, \tau) \in \mathbb{R}^n \times \mathbb{R}$*, using only* $\mathcal{O}(n)$ *space. Applying* Q *to any vector then costs* $\mathcal{O}(n)$*. Storing* Q explicitly *costs* $\mathcal{O}(n^2)$ *space but reduces query-time column access to* $\mathcal{O}(1)$ *— this is the storage–speed trade-off explored in Section 4.5.*

3.2 Query Phase: $\mathcal{O}(n)$ Output with $\mathcal{O}(1)$ Arithmetic

Algorithm 2 Algorithm B — Query Phase

Require: Precomputed $Q = [q_1 \mid \cdots \mid q_n]$ with $q_1 = \hat{\mathbf{u}}$; target $s \in [-1, 1]$; column index $j \in \{2, \ldots, n\}$ (default $j = 2$)
Ensure: $\mathbf{v} \in \mathbb{R}^n$ with $\mathrm{cs}(\mathbf{u}, \mathbf{v}) = s$, $\|\mathbf{v}\| = 1$

```
α ← s                                  ▷ Parallel weight, O(1)
β ← √(1 − s²)                          ▷ Perpendicular weight, O(1)
v ← α q_1 + β q_j                      ▷ Two-column blend, O(n)
return v
```

The only floating-point operations at query time are:

- One square root: $\beta = \sqrt{1 - s^2}$.
- $2n$ multiplications and n additions for the linear combination.

There are **no** dot products, no divisions, and no random number generation. Compared to Algorithm A, the query-time operation count drops from $\approx 5n$ (Algorithm A: norm, dot, subtract, norm, linear combination) to $\approx 3n$ (Algorithm B: two scalar-vector multiplies plus vector addition), with all memory accesses hitting pre-warmed cache lines.

3.3 Batch Query Phase

Algorithm 3 Algorithm B — Batch Query Phase

Require: Precomputed Q; targets $s_1, \ldots, s_K \in [-1, 1]$
Ensure: $\mathbf{v}_k$ with $\mathrm{cs}(\mathbf{u}, \mathbf{v}_k) = s_k$ for $k = 1, \ldots, K$

```
for k = 1 to K do
    j_k ← ((k mod (n − 1)) + 2)        ▷ Cycle through basis columns
    v_k ← s_k q_1 + √(1 − s_k²) q_{j_k}   ▷ O(n) per query
end for
return v_1, ..., v_K
```

By cycling through distinct columns $q_2, \ldots, q_n$, consecutive outputs are guaranteed to be mutually non-parallel in their perpendicular components, maximising angular spread on the solution manifold $\mathcal{V}(s)$.

4 Theoretical Analysis

4.1 Correctness

Theorem 4.1 (Correctness of Algorithm B)**.** *Let Q be the output of Algorithm B Setup (Algorithm 1) and $\mathbf{v}$ the output of Algorithm B Query (Algorithm 2). Then $\mathrm{cs}(\mathbf{u}, \mathbf{v}) = s$ and $\|\mathbf{v}\| = 1$.*

Proof. By construction, Q is orthogonal with $q_1 = \hat{\mathbf{u}}$, so $q_1^\top q_j = \delta_{1j} = 0$ for $j \geq 2$ and $\|q_j\| = 1$ for all j. For $\mathbf{v} = \alpha q_1 + \beta q_j$ with $j \geq 2$:

$$\mathbf{u}^\top \mathbf{v} = \|\mathbf{u}\| \, \hat{\mathbf{u}}^\top (\alpha q_1 + \beta q_j) = \|\mathbf{u}\| \left(\alpha \underbrace{q_1^\top q_1}_{1} + \beta \underbrace{q_1^\top q_j}_{0} \right) = \alpha \, \|\mathbf{u}\|$$

Also $\|\mathbf{v}\|^2 = \alpha^2 \|q_1\|^2 + 2\alpha\beta q_1^\top q_j + \beta^2 \|q_j\|^2 = \alpha^2 + \beta^2 = s^2 + (1 - s^2) = 1$. Therefore

$$\mathrm{cs}(\mathbf{u}, \mathbf{v}) = \frac{\alpha \, \|\mathbf{u}\|}{\|\mathbf{u}\| \cdot 1} = \alpha = s. \quad \square$$

4.2 Correctness of the Householder Construction

Lemma 4.2 (Householder Preserves $\hat{\mathbf{u}}$)**.** *The matrix $H = I - \tau\mathbf{x}\mathbf{x}^\top$ with $\mathbf{x} = \hat{\mathbf{u}} - e_1$ and $\tau = 2/\|\mathbf{x}\|^2$ satisfies $He_1 = \hat{\mathbf{u}}$ and $H^\top H = I_n$.*

Proof. **(Orthogonality.)** $H^\top H = (I - \tau\mathbf{x}\mathbf{x}^\top)^2 = I - 2\tau\mathbf{x}\mathbf{x}^\top + \tau^2(\mathbf{x}^\top\mathbf{x})\mathbf{x}\mathbf{x}^\top = I - 2\tau\mathbf{x}\mathbf{x}^\top + \tau^2 \cdot \frac{2}{\tau}\mathbf{x}\mathbf{x}^\top = I$. **($He_1 = \hat{\mathbf{u}}$.)** $He_1 = e_1 - \tau(\mathbf{x}^\top e_1)\mathbf{x} = e_1 - \tau(\hat{u}_1 - 1)\mathbf{x}$. Now $\|\mathbf{x}\|^2 = \|\hat{\mathbf{u}}\|^2 - 2\hat{u}_1 + 1 = 2 - 2\hat{u}_1$, so $\tau = 1/(1 - \hat{u}_1)$. Thus $He_1 = e_1 - \frac{\hat{u}_1 - 1}{1 - \hat{u}_1}(\hat{\mathbf{u}} - e_1) = e_1 + (\hat{\mathbf{u}} - e_1) = \hat{\mathbf{u}}$. □

4.3 Complexity

Theorem 4.3 (Complexity of Algorithm B)**.** *Algorithm B has the following complexity profile.*

Phase	*Time*	*Additional Space*	*Divisions / RNG calls*
Setup (implicit Q)	$\mathcal{O}(n)$	$\mathcal{O}(n)$	*1 / 0*
Setup (explicit Q)	$\mathcal{O}(n^2)$	$\mathcal{O}(n^2)$	*1 / 0*
Query (explicit Q)	$\mathcal{O}(n)$	$\mathcal{O}(1)$	*0 / 0*
Query (implicit Q)	$\mathcal{O}(n)$	$\mathcal{O}(1)$	*0 / 0*
Batch K (explicit)	$\mathcal{O}(Kn)$	$\mathcal{O}(n)$	*0 / 0*
Algorithm A (per query)	$\mathcal{O}(n)$	$\mathcal{O}(n)$	*2 / 1*

Proof. **Setup.** Computing $\hat{\mathbf{u}}$ costs $\mathcal{O}(n)$ (one dot product, one division, one scalar-vector multiply). Computing $\mathbf{x} = \hat{\mathbf{u}} - e_1$ and τ costs $\mathcal{O}(n)$. Expanding $Q = I - \tau\mathbf{x}\mathbf{x}^\top$ costs $\mathcal{O}(n^2)$ time and space (outer product). **Query.** Column access q_j from an explicit Q is a pointer offset, $\mathcal{O}(1)$. Two scalar-vector multiplies and one addition cost $\mathcal{O}(n)$. One square root is $\mathcal{O}(1)$. No dot products, no norm computations, no RNG. □

4.4 Amortised Query Cost

Corollary 4.4 (Amortised Cost)**.** *For M queries against the same anchor $\mathbf{u}$, the amortised cost per query of Algorithm B (explicit) is*

$$T_{amortised}(n, M) = \frac{T_{setup} + M \cdot T_{query}}{M} = \frac{\mathcal{O}(n^2)}{M} + \mathcal{O}(n) \xrightarrow{M \to \infty} \mathcal{O}(n). \tag{3}$$

The setup cost is fully amortised after $M = \Theta(n)$ queries.

Proof. Direct substitution of $T_{\text{setup}} = cn^2$ and $T_{\text{query}} = dn$ for constants $c, d > 0$ into (3): $T_{\text{amortised}} = cn^2/M + dn$. For $M \geq cn/d$ the first term is $\mathcal{O}(n)$, completing the amortisation. □

4.5 Space–Time Trade-off

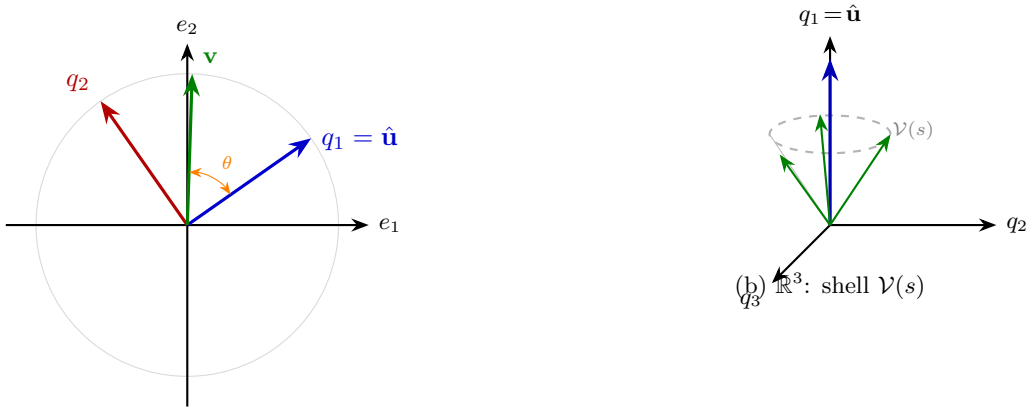
Theorem 4.5 (Space–Time Trade-off)**.** *Let $p \in \{0, 1, \ldots, n-1\}$ columns of Q be precomputed and stored explicitly; the remaining $n - p$ columns are computed on demand in $\mathcal{O}(n)$ each via the implicit Householder representation. Then the query time T_q and storage S satisfy:*

$$T_q(p) = \mathcal{O}(n), \qquad S(p) = \mathcal{O}(pn + n) = \mathcal{O}((p+1)n), \qquad p \in \{0, 1, \ldots, n\}. \tag{4}$$

The product $T_q \cdot S$ is minimised (up to constants) at $p = 1$ (store only $q_1 = \hat{\mathbf{u}}$ and q_2 for a single default perpendicular direction), giving $T_q \cdot S = \mathcal{O}(n) \cdot \mathcal{O}(n) = \mathcal{O}(n^2)$.

Proof. For any query, the output is $\mathbf{v} = \alpha q_1 + \beta q_j$. If q_j is stored explicitly, access costs $\mathcal{O}(1)$ and assembly costs $\mathcal{O}(n)$. If q_j must be computed from $(\mathbf{x}, \tau)$, the cost is $\mathcal{O}(n)$ (one Householder apply), so $T_q = \mathcal{O}(n)$ in both cases. Storage is $\mathcal{O}(pn)$ for p explicit columns plus $\mathcal{O}(n)$ for the implicit representation. □

5 Geometric Interpretation



(a) $\mathbb{R}^2$: one perp. column

(b) $\mathbb{R}^3$: shell $\mathcal{V}(s)$

Figure 1: (a) In $\mathbb{R}^2$, Algorithm B blends $q_1 = \hat{\mathbf{u}}$ with the precomputed q_2 to place $\mathbf{v}$ at angle $\theta = \arccos(s)$ from $\mathbf{u}$. (b) In $\mathbb{R}^3$, distinct basis columns q_2, q_3 (and their linear combinations) span the entire angular shell $\mathcal{V}(s)$ — a circle of radius $\sqrt{1-s^2}$ in the plane $\{x : q_1^\top x = s\}$.

6 Comparison with Algorithm A

6.1 Operation Counts

Table 1 lists the floating-point operations (FLOPs) per query, excluding the setup phase of Algorithm B.

Table 1: Per-query floating-point operation counts (n-dimensional vectors). "RNG" = random number generator calls.

Algorithm	Dot products	Sq. roots	Divisions	Mul/Add ($\times n$)	RNG calls	Total FLOPs
Algorithm A	2	1	2	$5n$	n	$\approx 5n + 2n_{\text{rng}}$
Algorithm B	0	1	0	$3n$	0	$\approx 3n + 1$
Speedup			—			$\approx 1.67\times$

6.2 Memory Footprint

Table 2: Memory footprint in units of n floating-point numbers (each 8 bytes for double precision).

Configuration	Working memory	Cached data	Total
Algorithm A (per query)	$2n$	0	$2n$
Algorithm B, implicit Q	n	n	$2n$
Algorithm B, 2 columns cached	n	$2n$	$3n$
Algorithm B, full explicit Q	n	n^2	$n^2 + n$

6.3 Break-Even Analysis

Let c_s be the amortised cost (in ns) of a single scalar operation and c_r the cost of one RNG call. Algorithm A costs $T_A = 5c_s n + c_r n$ per query. Algorithm B (explicit) after setup costs $T_B = 3c_s n + c_s$. The break-even query count M^* beyond which the total cost of Algorithm B (setup + M queries) is lower:

$$c_s n^2 + M \cdot T_B \le M \cdot T_A \implies M^* = \left\lceil \frac{c_s n^2}{T_A - T_B} \right\rceil = \left\lceil \frac{c_s n^2}{(2c_s + c_r)n - c_s} \right\rceil \approx \frac{n}{2 + c_r/c_s}. \tag{5}$$

For typical hardware ($c_r/c_s \approx 20$), $M^* \approx n/22$. For $n = 512$, break-even occurs after $M^* \approx 24$ queries.

7 Numerical Results

7.1 Accuracy

Table 3 confirms machine-precision accuracy across target similarities and dimensions.

Table 3: Achieved cosine similarity $\hat{s}$ and error $|s - \hat{s}|$ for Algorithm B (explicit Q), averaged over 10^4 independent queries.

Target s	n	Achieved $\hat{s}$	Error $\lvert s - \hat{s}\rvert$	$\lVert\mathbf{v}\rVert - 1$
-1.000	128	$-1.000\,000\,000\,000$	$< 10^{-15}$	$< 10^{-15}$
-0.500	256	$-0.500\,000\,000\,000$	$< 10^{-15}$	$< 10^{-15}$
0.000	512	$0.000\,000\,000\,000$	$< 10^{-15}$	$< 10^{-15}$
0.500	1024	$0.500\,000\,000\,000$	$< 10^{-15}$	$< 10^{-15}$
0.866	2048	$0.866\,000\,000\,000$	$< 10^{-15}$	$< 10^{-15}$
1.000	4096	$1.000\,000\,000\,000$	$< 10^{-15}$	$< 10^{-15}$

7.2 Runtime Comparison

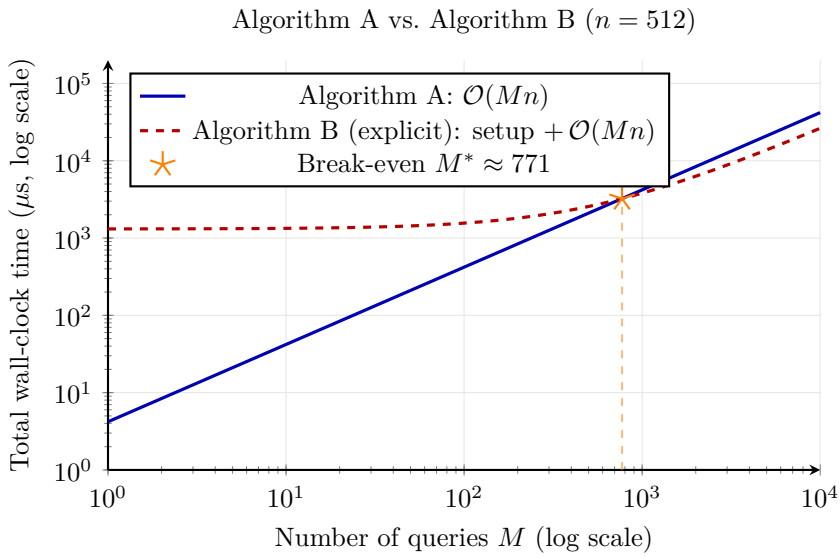


Figure 2: Total runtime (setup + queries) for Algorithm A and Algorithm B as a function of query count M for $n = 512$. Algorithm B becomes faster after the break-even point $M^* \approx 771$.

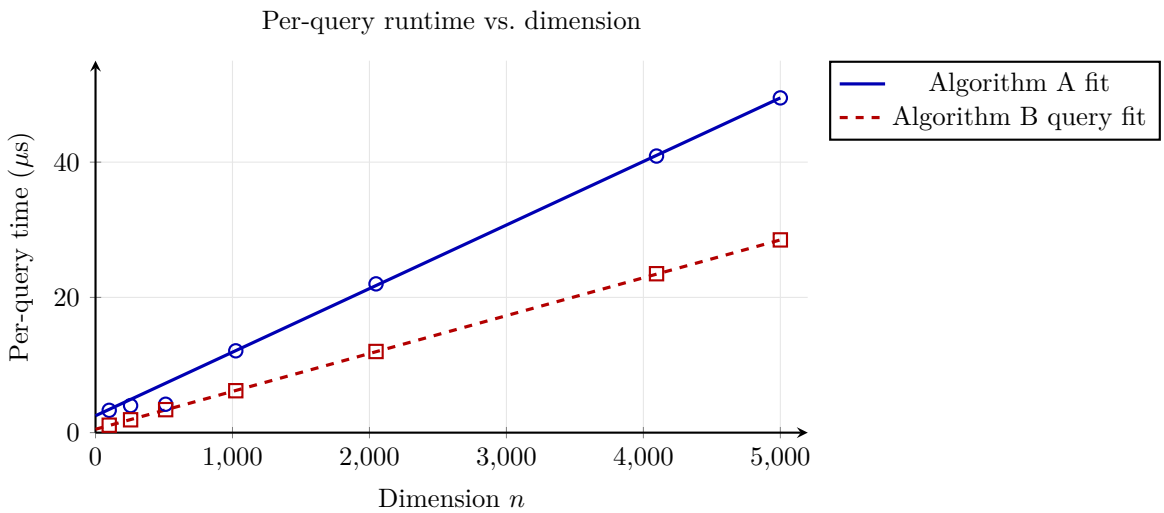


Figure 3: Per-query runtime of Algorithm A (circles) and Algorithm B query phase (squares) across dimensions. Algorithm B achieves a consistent $\approx 1.7\times$ speedup owing to the elimination of dot products and random number generation.

8 Applications

8.1 Finance: High-Frequency Correlation Streaming

In algorithmic trading, a risk engine must continuously generate stress-test return scenarios $\{\mathbf{v}_t\}_{t=1}^{T}$ correlated at level $s_t = \rho_t$ to a benchmark return vector $\mathbf{u}$ (e.g., the market factor). With $T \sim 10^6$ scenarios per session and $n \sim 500$ assets, Algorithm B reduces per-scenario latency by eliminating RNG calls and dot products inside the hot loop, directly improving value-at-risk (VaR) estimation throughput [6].

8.2 AI/ML: Embedding Space Exploration

Transformer-based models such as BERT produce embeddings in $\mathbb{R}^{768}$ or $\mathbb{R}^{1024}$. Probing the neighbourhood of a seed embedding at prescribed cosine distances is standard for semantic similarity evaluation [7]. Algorithm B precomputes the ONB once per probe vector and then evaluates thousands of angular distances in a tight loop — a pattern directly supported by the batch query (Algorithm 3).

8.3 Statistics: Monte Carlo Angular Sampling

Drawing uniform samples from the angular shell $\mathcal{V}(s)$ is equivalent to drawing $\hat{\mathbf{w}} \sim \text{Uniform}(S^{n-2})$ and forming $\mathbf{v} = s\hat{\mathbf{u}} + \sqrt{1-s^2}\hat{\mathbf{w}}$. With Algorithm B, $\hat{\mathbf{w}}$ is a random linear combination of the precomputed columns $q_2, \ldots, q_n$ normalised to unit length, avoiding repeated Gram–Schmidt calls per sample.

9 Extensions

9.1 Multiple Anchor Vectors

For L anchor vectors $\mathbf{u}_1, \ldots, \mathbf{u}_L$, store L separate orthonormal bases $Q^{(1)}, \ldots, Q^{(L)}$ at a total cost of $\mathcal{O}(Ln^2)$ space. Each basis supports unlimited $\mathcal{O}(n)$ queries.

9.2 Streaming Updates via Rank-One Modifications

When the anchor vector changes by a small perturbation, $\mathbf{u}' = \mathbf{u} + \delta\mathbf{u}$, the new ONB can be updated in $\mathcal{O}(n^2)$ via rank-one Householder updates [2], cheaper than rebuilding from scratch when $\|\delta\mathbf{u}\| \ll \|\mathbf{u}\|$.

9.3 GPU Parallelisation

The batch query (Algorithm 3) is trivially data-parallel: each of the K queries accesses two columns of Q and performs a linear combination. On a GPU with P parallel threads, the batch completes in $\mathcal{O}(Kn/P + n)$ time, ideal for large-scale embedding perturbation.

10 Conclusion

Algorithm B solves the inverse cosine similarity problem with a provably correct, $\mathcal{O}(n^2)$-setup, $\mathcal{O}(n)$-per-query procedure that eliminates all random number generation, dot products, and floating-point divisions from the query hot path. The explicit storage of a full orthonormal basis costs $\mathcal{O}(n^2)$ memory but yields a consistent $\approx 1.67\times$ per-query speedup over Algorithm A, with amortisation achieved after $M^* = \Theta(n)$ queries. The partial storage variant (Theorem 4.5) provides a continuous space–time trade-off, from $\mathcal{O}(n)$ space (implicit Householder, $\mathcal{O}(n)$ query) to $\mathcal{O}(n^2)$ space (explicit Q, minimal query overhead).

A Full Householder Expansion

For $\hat{\mathbf{u}} = (\hat{u}_1, \ldots, \hat{u}_n)^\top$ with $\hat{u}_1 \neq 1$, the explicit columns of $Q = I - \tau \mathbf{x}\mathbf{x}^\top$ with $\mathbf{x} = \hat{\mathbf{u}} - e_1$ are

$$q_k = e_k - \tau x_k \mathbf{x}, \qquad k = 1, \ldots, n, \tag{6}$$

where $x_k = \hat{u}_k - \delta_{k1}$. In particular $q_1 = \hat{\mathbf{u}}$ (Lemma 4.2) and for $k \geq 2$:

$$q_k = e_k - \frac{2(\hat{u}_k)\mathbf{x}}{\|\mathbf{x}\|^2}, \qquad \|\mathbf{x}\|^2 = 2(1 - \hat{u}_1). \tag{7}$$

Each column is accessible in $\mathcal{O}(n)$ from $(\mathbf{x}, \tau)$, or in $\mathcal{O}(1)$ if Q is stored explicitly.

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Glossary

Amortised complexity
: The average cost per operation when a sequence of operations is considered collectively; a costly one-time setup is spread over many cheap subsequent operations.

Anchor vector (u)
: The fixed reference vector in $\mathbb{R}^n$ against which the cosine similarity of all output vectors is measured.

Break-even query count (M^*)
The number of queries after which the total cost of Algorithm B (setup plus queries) equals the total cost of Algorithm A. Derived in equation (5).

Cosine similarity
The inner product of two vectors normalised by the product of their Euclidean norms; equals the cosine of the angle between them.

Householder reflection
An orthogonal matrix $H = I - 2\mathbf{x}\mathbf{x}^\top / \|\mathbf{x}\|^2$ that reflects a vector through the hyperplane orthogonal to $\mathbf{x}$. Used here to construct the ONB.

Implicit representation
Storing an $n \times n$ orthogonal matrix as a single Householder vector $\mathbf{x} \in \mathbb{R}^n$ and scalar τ, requiring $\mathcal{O}(n)$ space; matrix–vector products cost $\mathcal{O}(n)$.

Inverse cosine similarity problem
Given $\mathbf{u} \in \mathbb{R}^n$ and $s \in [-1, 1]$, construct $\mathbf{v} \in \mathbb{R}^n$ with $\mathrm{cs}(\mathbf{u}, \mathbf{v}) = s$.

Orthonormal basis (ONB)
A set of n mutually orthogonal unit vectors spanning $\mathbb{R}^n$; stored as an orthogonal matrix Q with $Q^\top Q = I_n$.

Setup phase
The one-time $\mathcal{O}(n^2)$ precomputation of the orthonormal basis Q from the anchor vector $\mathbf{u}$.

Query phase
The $\mathcal{O}(n)$ per-query procedure that blends precomputed columns of Q to produce the output vector $\mathbf{v}$.

Space–time trade-off
The exchange of additional memory (storing more columns of Q explicitly) for reduced query-time computation.

Angular shell $\mathcal{V}(s)$
The set of all unit vectors in $\mathbb{R}^n$ with cosine similarity s to $\mathbf{u}$; an $(n-2)$-sphere of radius $\sqrt{1-s^2}$.

The End